Dual-Control Bionic Hand System

Comprehensive Technical Documentation & Circuit Schematics

Actuation:	5x MG90S Metal-Gear Micro Servos	**Control Mode 1:**	Computer Vision (OpenCV + MediaPipe)
Structure:	3D Printed Anthropomorphic Hand & Wrist	**Control Mode 2:**	Glove-Based Flex Resistive Sensors

1. System Architecture & Overview

This document presents the complete engineering documentation and circuit designs for the 3D-printed Bionic Hand assembly. The system employs tendon-driven mechanical actuation across five independent finger digits (Thumb, Index, Middle, Ring, Pinky), controlled by five high-torque **MG90S metal-gear micro servos** mounted inside the forearm structure.

The robotic hand features a dual-mode control pipeline:

1. **Vision-Controlled Mode (OpenCV / MediaPipe):** Real-time contactless gesture tracking using a camera, processing hand landmark coordinates via Python, and streaming actuation angles to the microcontroller via Serial communication.
2. **Haptic/Glove-Controlled Mode (Flex Sensors):** Direct anatomical mirroring via a sensor glove equipped with 5 flex sensors forming voltage divider circuits read by microcontroller ADC channels.

2. Hardware Components & Specifications

Component	Quantity / Specification	Functional Purpose
Microcontroller	Arduino Uno / Nano (ATmega328P)	Processes ADC flex inputs or UART commands to output PWM servo signals.
Servo Actuators	5x MG90S (Metal Gear, 1.8–2.2 kg·cm)	Pulls tension cables/strings for flexion; elastic recoil handles extension.
Flex Sensors	5x 2.2" Flexible Resistive Sensors (~10k–30kΩ)	Detects finger bending on the user's control glove.
Resistors	5x 10kΩ Fixed Resistors (¼W)	Voltage divider pair for flex sensor analog voltage conversion.
Power Supply	External 5V DC (3A–5A High Current)	Drives 5x MG90S servos under peak stall currents (prevents MCU brownout).
Decoupling Capacitor	1x 1000μF / 16V Electrolytic	Filters voltage spikes during sudden servo motor draw.

3. Circuit Schematics & Wiring Connections

3.1 Microcontroller Pinout Mapping Table

Hand Digit	Servo Signal Pin (PWM)	Flex Sensor Input Pin (ADC)	Voltage Divider Resistor
Thumb	Digital Pin D3	Analog Pin A0	10 kΩ to GND
Index Finger	Digital Pin D5	Analog Pin A1	10 kΩ to GND
Middle Finger	Digital Pin D6	Analog Pin A2	10 kΩ to GND
Ring Finger	Digital Pin D9	Analog Pin A3	10 kΩ to GND
Pinky Finger	Digital Pin D10	Analog Pin A4	10 kΩ to GND

3.2 Complete Circuit Block Diagram

```
+------------------------------------------------------------------------------+
```

```
|                               POWER SUPPLY (5V DC 3A-5A)                            |
|  +5V -----------------------+----------------------+-------------------+           |
|                             |                      |                   |           |
|  GND -----------+-----------|----------------------|-------------------|-- (MCU)   |
|                 |           |                      |                   |           |
|            [1000uF Cap]     |                      |                   |           |
+-----------------|-----------|----------------------|-------------------|-----------+
                  |           |                      |                   |
                  |           v                      |                   v
     +------------|-----------+----------+           |      +------------+-----------+
     |     5x MG90S SERVO MOTORS          |           |      |  5x FLEX SENSORS (GLOVE)   |
     |  Red (+5V)  <-- Common 5V Bus      |           |      | +5V Bus --> Terminal 1    |
     |  Black(GND) <-- Common GND Bus     |           |      |                            |
     |  Yellow/Orange (Signal):           |           |      | Terminal 2 (Analog Out):   |
     |   - Servo 1 (Thumb)   --> Pin D3   |           |      |  - Flex 1 (Thumb)  --> A0  |
     |   - Servo 2 (Index)   --> Pin D5   |           |      |  - Flex 2 (Index)  --> A1  |
     |   - Servo 3 (Middle)  --> Pin D6   |           |      |  - Flex 3 (Middle) --> A2  |
     |   - Servo 4 (Ring)    --> Pin D9   |           |      |  - Flex 4 (Ring)    --> A3  |
     |   - Servo 5 (Pinky)   --> Pin D10  |           |      |  - Flex 5 (Pinky)  --> A4  |
     +------------------------------------+           |      +------------+-----------+
                                                      |                   |
                                                      +--- [ 10k Resistor ] ---> GND
```

Critical Wiring Rule

Common Ground Connection: Always connect the External 5V Power Supply GND to the Arduino GND pin. Without a common ground reference, PWM signals will fluctuate, causing severe servo jitter and unresponsiveness. Never power 5 servos directly from the Arduino 5V pin!

4. Implementation Code

4.1 Control Mode 1: OpenCV + MediaPipe Python Script (PC Side)

```python
import cv2
import mediapipe as mp
import serial
import time

# Initialize Serial Communication with Arduino
ser = serial.Serial('COM3', 9600, timeout=1)
time.sleep(2)

mp_hands = mp.solutions.hands
hands = mp_hands.Hands(max_num_hands=1, min_detection_confidence=0.7)
cap = cv2.VideoCapture(0)

def get_angle(tip, mcp):
    # Simple landmark distance threshold for finger bending
    return 180 if tip.y > mcp.y else 0

while cap.isOpened():
    ret, frame = cap.read()
    if not ret: break
    frame = cv2.flip(frame, 1)
    rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    res = hands.process(rgb)

    if res.multi_hand_landmarks:
        for hand_lms in res.multi_hand_landmarks:
            lm = hand_lms.landmark
            # Calculate state for 5 digits (Thumb, Index, Middle, Ring, Pinky)
            angles = [
                180 if lm[4].x < lm[3].x else 0,    # Thumb
                get_angle(lm[8], lm[6]),          # Index
                get_angle(lm[12], lm[10]),        # Middle
```

```
                get_angle(lm[16], lm[14]),       # Ring
                get_angle(lm[20], lm[18])        # Pinky
            ]
            cmd = f"{angles[0]},{angles[1]},{angles[2]},{angles[3]},{angles[4]}
"
            ser.write(cmd.encode())

    cv2.imshow("Bionic Hand Tracker", frame)
    if cv2.waitKey(1) == ord('q'): break

cap.release()
cv2.destroyAllWindows()
```

4.2 Control Mode 2: Flex Sensor Direct Microcontroller Firmware (Arduino C++)

```
#include <Servo.h>

Servo servos[5];
const int servoPins[5] = {3, 5, 6, 9, 10};
const int flexPins[5]  = {A0, A1, A2, A3, A4};

// Calibration min/max ADC values for Flex Sensors
int flexMin[5] = {300, 300, 300, 300, 300};
int flexMax[5] = {700, 700, 700, 700, 700};

void setup() {
  for (int i = 0; i < 5; i++) {
    servos[i].attach(servoPins[i]);
  }
}

void loop() {
  for (int i = 0; i < 5; i++) {
    int flexVal = analogRead(flexPins[i]);
    // Map analog reading to 0-180 degree servo sweep
    int angle = map(flexVal, flexMin[i], flexMax[i], 0, 180);
    angle = constrain(angle, 0, 180);
    servos[i].write(angle);
  }
  delay(20);
}
```

5. Calibration & Testing Guidelines

- **Mechanical Tensioning:** Adjust nylon string tension in the 3D-printed channels so that 0° fully extends fingers and 180° provides full palm closure without stalling MG90S motors.
- **ADC Thresholding:** Measure minimum (flat) and maximum (bent) ADC values for each flex sensor using `Serial.println(analogRead(A0))` and update `flexMin` / `flexMax` arrays.